

AGILE DEVELOPMENT AND DEVOPS

LAB FAT

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Online Auction System

Problem Statement:

Ensure bids are recorded and highest bidder is selected correctly.

Tasks:

1. Develop bidding logic
2. Write JUnit tests for bid validation
3. Implement CI/CD pipeline with automated testing

Set up version control using GitHub

Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits

Automate build process using Maven

Integrate automated unit tests in pipeline

Configure CD pipeline for automatic deployment to staging/production

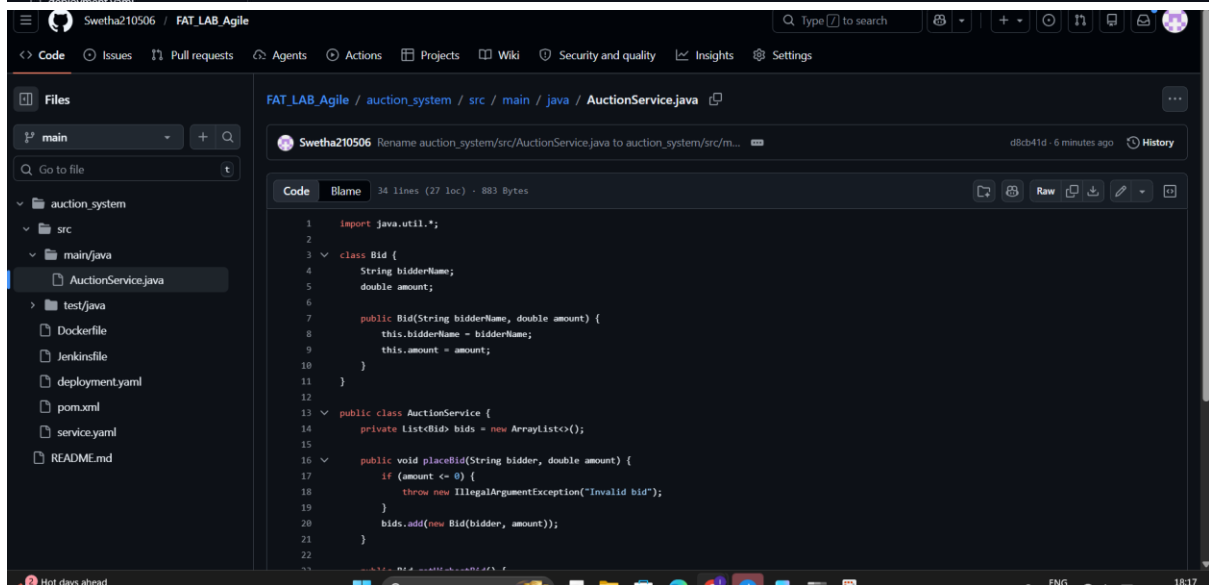
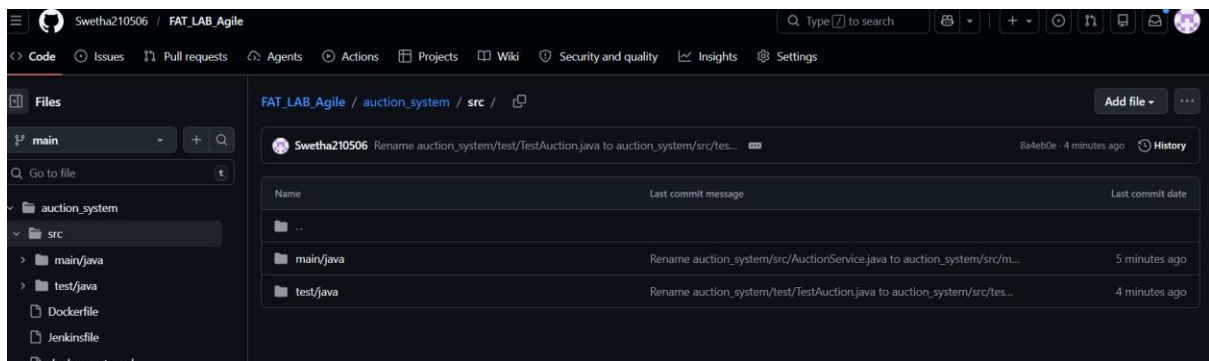
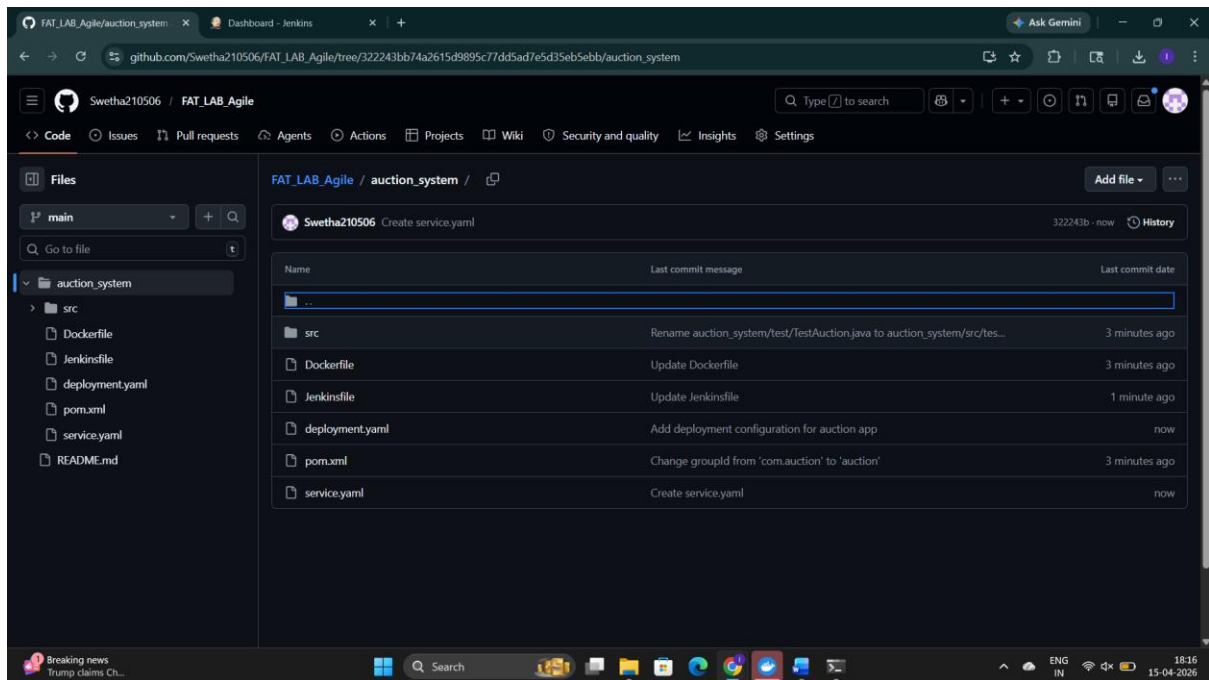
4. Deploy services on Docker and Kubernetes

Containerize application using Docker (write Dockerfile)

➤ Build and push Docker image to container registry (Docker Hub)

Create Kubernetes deployment and service YAML files.

Github :



FAT_LAB_Agile/auction_system x Dashboard - Jenkins x + Ask Gemini

github.com/Swetha210506/FAT_LAB_Agile/blob/main/auction_system/pom.xml

Files

- main
- Go to file
- t
- main
- src
- main/java
 - AuctionService.java
- test/java
- Dockerfile
- Jenkinsfile
- deployment.yaml
- pom.xml
- service.yaml
- README.md

FAT_LAB_Agile / auction_system / pom.xml

Swetha210506 Change groupid from 'com.auction' to 'auction' 5113aa6 · 9 minutes ago History

Code Blame 21 lines (18 loc) · 553 Bytes

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0">
2   <modelVersion>4.0.0</modelVersion>
3
4   <groupId>auction</groupId>
5   <artifactId>auction-system</artifactId>
6   <version>1.0</version>
7
8   <properties>
9     <maven.compiler.source>17</maven.compiler.source>
10    <maven.compiler.target>17</maven.compiler.target>
11  </properties>
12
13  <dependencies>
14    <dependency>
15      <groupId>org.junit.jupiter</groupId>
16      <artifactId>junit-jupiter</artifactId>
17      <version>5.9.0</version>
18      <scope>test</scope>
19    </dependency>
20  </dependencies>
21 </project>
```

FAT_LAB_Agile/auction_system x #5 - FAT_LAB_Agile - Jenkins x + Ask Gemini

github.com/Swetha210506/FAT_LAB_Agile/blob/main/auction_system/deployment.yaml

Swetha210506 / FAT_LAB_Agile

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Files

- main
- Go to file
- t
- main
- src
- Dockerfile
- Jenkinsfile
- deployment.yaml
- pom.xml
- service.yaml
- README.md

FAT_LAB_Agile / auction_system / deployment.yaml

Swetha210506 Add deployment configuration for auction app f22e2af · 27 minutes ago History

Code Blame 19 lines (19 loc) · 342 Bytes

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: auction-app
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: auction
10  template:
11    metadata:
12      labels:
13        app: auction
14    spec:
15      containers:
16        - name: auction-container
17          image: yourdockerhub/auction-app
18          ports:
19            - containerPort: 8080
```

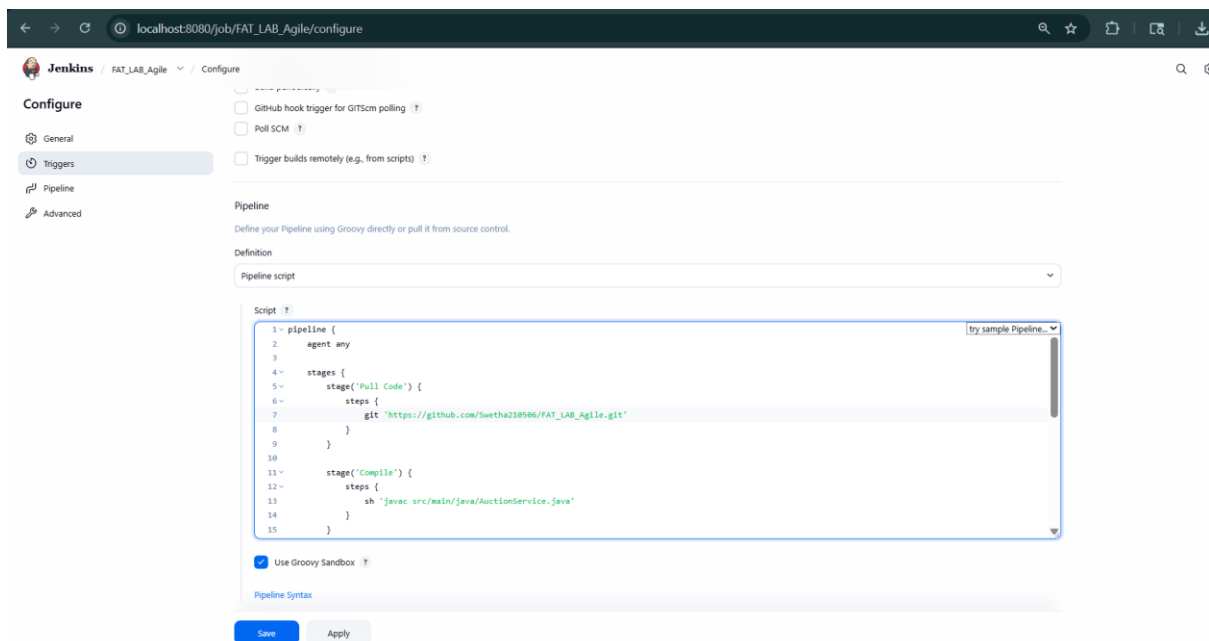
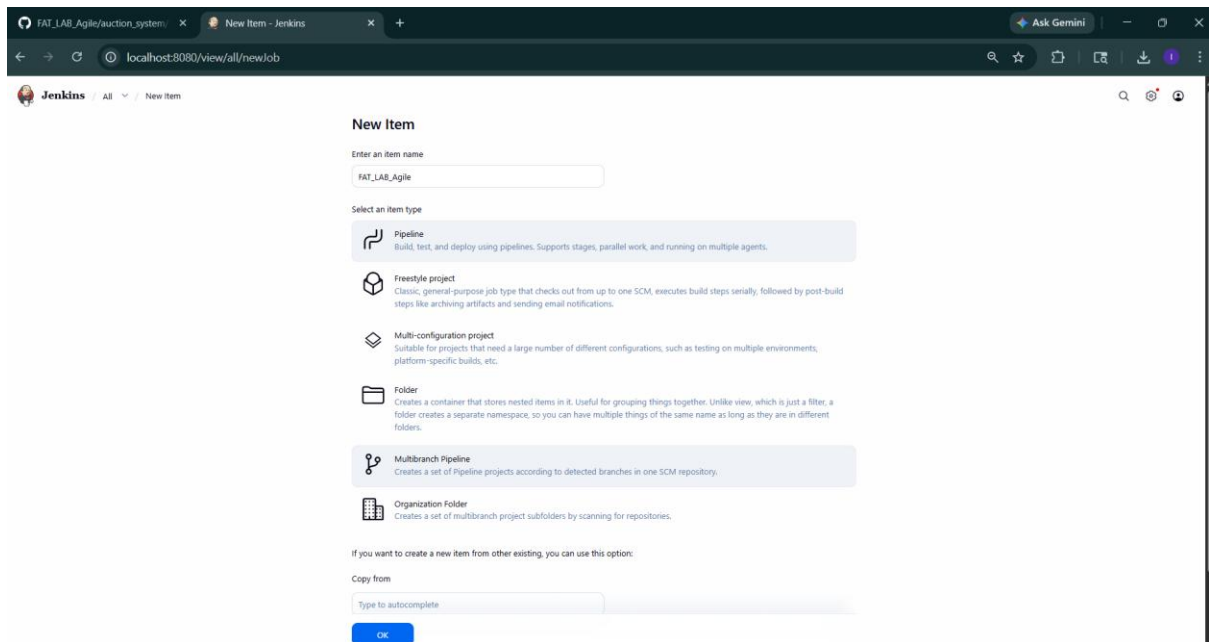
FAT_LAB_Agile / auction_system / service.yaml

Swetha210506 Create service.yaml 322243b · 27 minutes ago History

Code Blame 12 lines (12 loc) · 184 Bytes

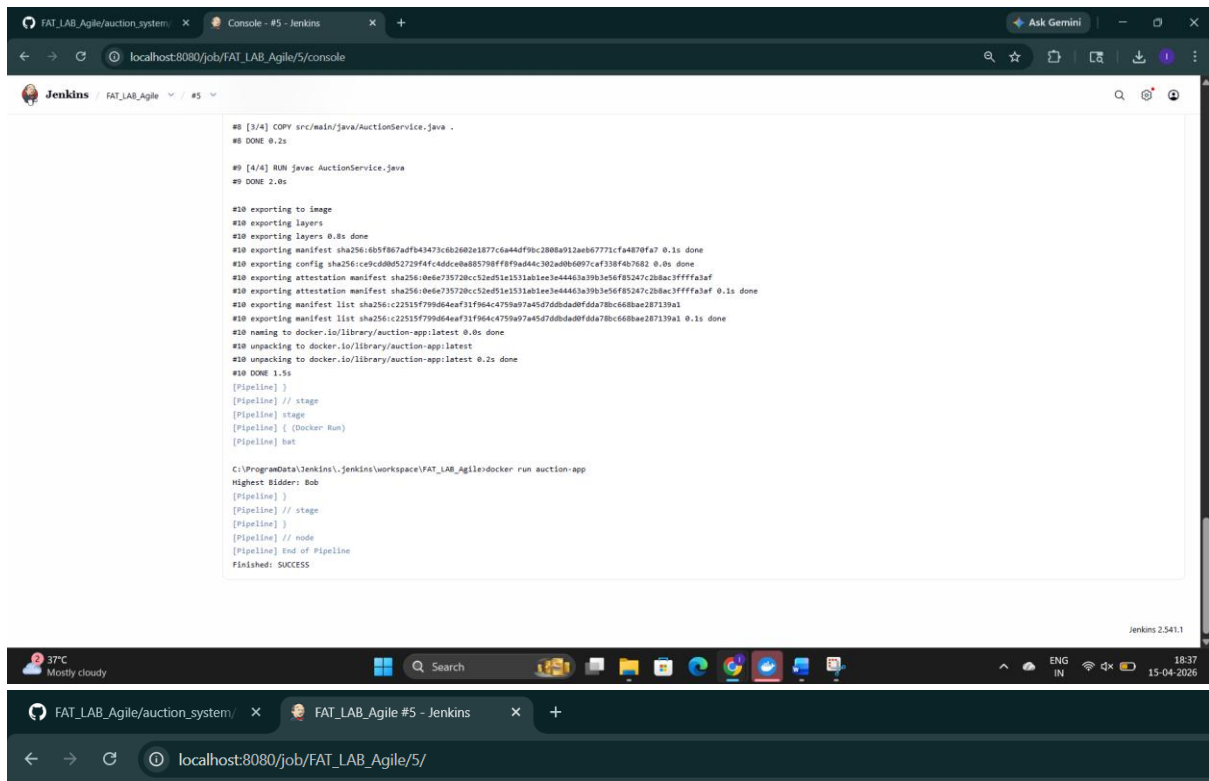
```
1 apiVersion: v1
2 kind: Service
3 metadata:
4   name: auction-service
5 spec:
6   type: NodePort
7   selector:
8     app: auction
9   ports:
10     - port: 80
11       targetPort: 8080
12       nodePort: 30007
```

Jenkins:



Output:

37°C Mostly cloudy Search 18:36 15-04-2026



Pipeline script:

```
pipeline {
```

```
    agent any
```

```
    stages {
```

```
        stage('Pull Code') {
```

```
            steps {
```

```
    git branch: 'main', url:  
'https://github.com/Swetha210506/FAT_LAB_Agile.git'
```

```
    }  
}
```

```
stage('Compile') {  
    steps {  
        bat 'javac auction_system/src/main/java/AuctionService.java'  
    }  
}
```

```
stage('Run') {  
    steps {  
        bat 'java -cp auction_system/src/main/java AuctionService'  
    }  
}
```

```
stage('Docker Build') {  
    steps {  
        bat 'docker build -t auction-app auction_system'  
    }  
}
```

```
stage('Docker Run') {  
    steps {  
        bat 'docker run auction-app'
```

```
}  
  
}  
  
}  
  
}
```

The screenshot shows the Jenkins web interface in a browser. The URL is `localhost:8080/job/FAT_LAB_Agile/5/stages/`. The pipeline is named `#5` and is in a successful state. The build graph shows the following stages: `Start`, `Pull Code`, `Compile`, `Run`, `Docker Build`, `Docker Run`, and `End`. All stages are marked with green checkmarks, indicating success. The `Docker Run` stage is expanded, showing a `Windows Batch Script` with the command `docker run auction-app`. The build log shows the output of the command, including the path `C:\ProgramData\Jenkins\jenkins\workspace\FAT_LAB_Agile\docker run auction-app` and the highest bidder `Bob`. The Jenkins version is `2.541.1`.

Dockers :

The screenshot shows the Docker Desktop interface. The left sidebar contains navigation options: `Containers`, `Images`, `Volumes`, `Kubernetes`, `Builds`, `Models`, `MCP Toolkit`, `Docker Hub`, `Docker Scout`, and `Extensions`. The main panel displays the `Images / auction-app:latest` image. The image is created 2 minutes ago and has a size of 635.96 MB. The image is analyzed by Docker Scout. The layers of the image are listed in a table:

Layer ID	Layer Description	Size
0	ARG RELEASE	0 B
1	ARG LAUNCHPAD_BUILD_ARCH	0 B
2	LABEL org.opencontainers.image.version=24.04	0 B
3	ADD file:0f6466425c4f1800ase9224ddc3437b90c829cea...	87.6 MB
4	CMD ["/bin/bash"]	0 B
5	ENV JAVA_HOME=/opt/java/openjdk	0 B
6	ENV PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/lo...	0 B
7	ENV LANG=en_US.UTF-8 LANGUAGE=en_US LC_ALL=en...	0 B
8	RUN /bin/sh -c set -eux; apt-get update; DEBIAN_FRONTEN...	68.82 MB
9	ENV JAVA_VERSION=jdk-17.0.18+8	0 B
10	RUN /bin/sh -c set -eux; ARCH=\$(dpkg --get-architecture)	281.13 MB

The image has not been analyzed by Docker Scout. A message states: "This image has not been analyzed. You can use Docker Scout to analyze local images and list its vulnerabilities." A `Start analysis` button is available. The bottom status bar shows the engine is running, Kubernetes is starting, and the system has 1.15 GB RAM, 0.63% CPU, and 4.16 GB disk used. The system clock shows 18:39 on 15-04-2026.


```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.048 s -- in AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\balag\test-java-app\target\test-java-app-1.0.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.768 s

```

Git:

The screenshot shows an IDE with the following components:

- EXPLORER:** Shows the project structure for 'FAT_LAB_Agile' with folders like 'src' and 'test', and files like 'AuctionService.java' and 'AuctionServiceTest.java'.
- EDITOR:** Displays the code for 'AuctionService.java'. It includes a 'Bid' class with attributes 'bidderName' and 'amount', and an 'AuctionService' class with methods 'placeBid', 'getHighestBid', and a private 'bids' list.
- TERMINAL:** Shows the execution of 'git commit' and 'git status' commands. Both commands result in 'nothing to commit, working tree clean'.

```

1  import java.util.*;
2
3  class Bid {
4      String bidderName;
5      double amount;
6
7      public Bid(String bidderName, double amount) {
8          this.bidderName = bidderName;
9          this.amount = amount;
10     }
11 }
12
13 public class AuctionService {
14     private List<Bid> bids = new ArrayList<>();
15
16     public void placeBid(String bidder, double amount) {
17         if (amount <= 0) {
18             throw new IllegalArgumentException("Invalid bid");
19         }
20         bids.add(new Bid(bidder, amount));
21     }
22
23     public Bid getHighestBid() {
24         return Collections.max(bids, Comparator.comparingDouble(b -> b.amount));
25     }
26 }

```

```

PS C:\OneDrive\Desktop\CLG\FAT\FAT_LAB_Agile> git commit
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

PS C:\OneDrive\Desktop\CLG\FAT\FAT_LAB_Agile> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

```